

Fabrizio S. Lozano Negro

PERSONAL DATA

Place and Date of Birth:	Rafaela, Argentina January 21st 1995.
Resident status:	Argentine resident.
Address:	25 de Mayo 1350, Cordoba, Argentina.
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EDUCATION

2019-2024	PhD in Physics, Instituto de Física Enrique Gaviola - Universidad Nacional de Cordoba, Cordoba, Argentina. Thesis: “Ecos de Loschmidt en el estudio de la Dinámica Cuántica y Decoherencia en excitaciones de carga y espín en sistemas complejos observados por Resonancia Magnética Nuclear.” (Loschmidt echoes in the study of Quantum Dynamics and Decoherence in charge and spin excitations in complex systems observed by Nuclear Magnetic Resonance.) Advisors: Dr. Horacio M. Pastawski, Dr. Gonzalo A. Álvarez.
2017-2018	Master Degree in Physics, Instituto Balseiro - Universidad Nacional de Cuyo, Bariloche, Argentina Courses Passed: 6. Overall Score: 9 over 10. Thesis: “Monitoreando la dinámica de muchos espines con ecos de reversión temporal por Resonancia Magnética Nuclear.” (Monitoring many-spin dynamics with time reversal echoes by NMR) Advisors: Dr. Gonzalo A. Alvarez, Dra. Analía Zwick.
2015-2017	“Licenciatura en Física” (Equivalent to Bachelor in Physics) at Instituto Balseiro - Universidad Nacional de Cuyo, Bariloche, Argentina. Courses Passed: 19. Overall Score: 9.16 over 10. Thesis: “Efectos de localización e irreversibilidad cuántica con técnicas de Resonancia Magnética Nuclear.” (Effects of quantum localisation and irreversibility with NMR techniques) Advisors: Dr. Gonzalo A. Alvarez, Dra. Analía Zwick.
2013-2015	First two years of Lic. en Física at Facultad de Ciencias Exactas, Ingeniería y Agrimensura - Universidad Nacional de Rosario, Rosario, Argentina. Courses Passed: 13. Overall Score: 9.46 over 10.

RESEARCH POSITIONS

2024-Present	Postdoctoral fellow, University of Florence. Project APACE .
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PUBLICATIONS

2024	“Global Out of Time Order Correlators as a Signature of Scrambling Dynamics of Local Observables” Lozano-Negro F.S., Sánchez C.M., Chattah A.K., Álvarez G.A., Pastawski H.M. Phys. Rev. A 110, 042410
2024	“Universal stability of coherently diffusive 1D systems with respect to decoherence.” Lozano-Negro F.S., Alvarez Navarro E., Chávez Calderon N., Mattiotti F., Borgonovi F., Pastawski H.M., Celardo G.L. Phys. Rev. A 109, 042213.
2022	“Simulating a catalyst induced quantum dynamical phase transition of a Heyrovsky reaction with different models for the environment.” Lozano-Negro F.S., Ferreyra-Ortega M.A., Bendersky D., Fernández-Alcázar L., and Pastawski H.M. J. Phys.: Condens. Matter 34 214006.
2021	“Ergodicity breaking in an incommensurate system observed by OTOCs and Loschmidt Echoes: From quantum diffusion to sub-diffusion.” - Lozano-Negro F. S., Zangara P., Pastawski H. M.. Chaos, Solitons & Fractals 150, 111175.

TEACHING EXPERIENCE

2024	Assistant professor. Universidad nacional de Córdoba. Course: “Discrete Mathematics I”.
2023	Assistant professor. Universidad nacional de Córdoba. Course: “Algebra I”.
2021-2025	Assistant professor. Universidad Favaloro. Course: “Physics IV”.
2020	Assistant professor. Universidad nacional de Cordoba. Course: “Physics I”.

POSTGRADUATE COURSES

2023	School on Quantum Many-Body Phenomena out of Equilibrium: from Chaos to Criticality - ICTP - Trieste, Italy.
2023	Conference and Advanced School on Low-Dimensional Quantum Systems - ICTP - UCh, Chile.
2022	Probability and Stochastic processes. - C. Briozzo (FaMAF)
2022	Computational Physics. V. Marconi & A. Banchio (FaMAF)
2020	Molecular Electronic. - H. Pastawski (FaMAF)
2019	Introduction to university reaching. - Cristina Esteley, Enrique Coleoni (FaMAF)
2019	XXI Giambiagi Winter School - Buenos Aires (Quantum simulations and quantum metrology with cold trapped ions).
2019	Methods for signal hyperpolarization and preservation methods in Nuclear Magnetic Resonance. -B. Franzoni (FaMAF-UNC).
2019	Quantum field theory in solids: An introduction to the Many-Body Problem. - H. Pastawski (FaMAF-UNC).
2018	Solid State I - P. Cornaglia (Instituto Balseiro).
2018	Summer School on Collective Behaviour in Quantum Matter -ICTP.
2018	Neuronal Networks - G. Mato (Instituto Balseiro).
2018	Phase Transitions - E. Jagla (Instituto Balseiro).
2018	Materials characterisation - A. Fainstein (Instituto Balseiro).
2017	Disorder - M. O. Cáceres (Instituto Balseiro).
2017	Phenomenology in Condensed matter - J. Sereni & J. Guimpel (Instituto Balseiro).

PRESENTATIONS IN CONGRESSES

- 2026 | APACE Biannual Spring Meeting 2026, Heriot-Watt University, Edinburgh, United Kingdom.
- 2025 | APACE Biannual Autumn Meeting 2025, Institute of Physical Chemistry (ICHO PAN), Warsaw, Poland.
- 2025 | APACE Biannual Spring Meeting 2025, Max Planck Institute for the Science of Light (MPG), Erlangen, Germany.
- 2024 | “Global Out of Time Order Correlators (OTOCs) as a signature of the scrambling of local observables.” Talk
“Global OTOCs as sensors of the correlation of local observables.” Poster
109th Meeting of the Asociación Física Argentina + CUANTOS 6 - San Luis, Argentina.
- 2024 | 104th IUVESTA Workshop: Current trends in low-dimensional materials: From molecular systems to heterostructured nanomaterials.
- 2024 | “Universal stability of coherently diffusive 1D systems with respect to decoherence.” Talk. San Luis, Argentina.
- 2023 | “Universal robustness to decoherence in the quantum diffusive regime.” Poster - *ICTP smr 3867* - Trieste, Italy.
- 2023 | “Decoherence-independent transport in the Harper-Hofstadter-Aubry-André model at criticality.” Poster - *ICTP smr 3820* - Santiago de Chile, Chile.
- 2023 | “Dephasing independent transport in the diffusive coherent regime around criticality.” Poster - *Quantos 5* - FaMAF- Cordoba, Argentina.
- 2022 | “Ergodicity breaking in an incommensurate system observed by OTOCs and Loschmidt Echoes: From quantum diffusion to subdiffusion.” Poster - *107^o RAFA* - Bariloche, Argentina.
- 2021 | “Ergodicity breaking in an incommensurate system observed by OTOCs and Loschmidt Echoes: From quantum diffusion to sub-diffusionn.” Poster - *PROTOCOL21*, Max Planck Institute, Dresden, Germany.
- 2019 | “Linking localization effects on the dynamic behavior of many-spin systems with quantum irreversibility.” Lozano-Negro F. S., Dall’Alba A., Zwick A., Alvarez G. A.. Poster - “*Giambiagi Winter School*”, Buenos Aires, Argentina.
- 2018 | “Relacionando efectos de localización en el comportamiento dinámico de muchos espines con irreversibilidad cuántica.” Lozano-Negro F. S., Dall’Alba A., Zwick A., Alvarez G. A. . Poster “*103^o RAFA*”, Buenos Aires, Argentina.
- 2018 | “Linking localization effects on the dynamic behavior of many-spin systems with quantum irreversibility.” Lozano-Negro F. S., Dall’Alba A., Zwick A., Alvarez G. A.. Poster - “*Summer School on Collective Behaviour in Quantum Matter*”, Trieste, Italy
- 2017 | “Analysis of the movement and electrical properties of cells grown in vitro using a new measurement technique with lock-in amplifiers.” Lozano-Negro F. S., Bonetto F. ,Bellotti M., Giana F. . Poster presentado en “*102^o RAFA*”, La Plata, Argentina

SEMINARS AND TALKS

- 2023 | “Universal stability of coherently diffusive 1D systems with respect to decoherence.” Invited seminar - MIT - Boston, USA.
- 2023 | “Universal stability of coherently diffusive 1D systems with respect to decoherence.” Invited seminar - UCSC - Brescia, Italy.
- 2023 | “Monitoreo cuántico en células vivas: detectando radicales libres.” (*Quantum monitoring of living cells: detecting free radicals.*) Physics’s seminars - FaMAF - Cordoba, Argentina.
- 2020 | “Transición de localización en excitaciones multiespín en un potencial de Harper-Hofstadter-Aubry-Andre evaluadas por OTOCs y Ecos de Loschmidt.” (*Localization transition in multispin excitation with a Harper-Hofstadter-Aubry-Andre potential evaluated through OTOCs and Lochmidt Echoes.*) 105º RAFA - FaMAF - Cordoba, Argentina.
- 2018 | “Monitoreando la dinámica de muchos espines con ecos de reversión temporal por Resonancia Magnetica Nuclear” Master Thesis - Instituto Balserio, Bariloche, Argentina.
- 2018 | “Relacionando efectos de localización en el comportamiento dinámico de muchos espines con irreversibilidad cuántica.” 103 RAFA - UBA -Buenos Aires, Argentina.
- 2017 | “Efectos de localización e irreversibilidad cuántica con técnicas de resonancia magnética nuclear” Thesis - Instituto Balserio, Bariloche, Argentina.
- 2017 | “Análisis del movimiento y propiedades eléctricas de células cultivadas in vitro utilizando una nueva técnica de medición con lock-in” Laboratorio avanzado - Instituto Balserio, Bariloche, Argentina.
- 2014 | “Una charla sobre huevos”. Outreach talk at “Jornadas de Física” Fernandez L. & Lozano-Negro F. S. - UNR, Rosario, Argentina.

HONOURS AND AWARDS

- 2022 | Special prize for the best introduction of the papers in “[Special issue: the physics of electrocatalysis](#)”.
- 2017 | Highest overall score of the 61st promotion of Licenciados en Física.